

Analysis 2026

List 2

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Exercise 1 (Tao 3.3.2).

Exercise 2 (Tao 3.3.3).

Exercise 3 (Tao 3.3.4).

Exercise 4 (Tao 3.3.5).

Exercise 5 (Tao 3.3.7).

Exercise 6.

- a) Let $f : X \rightarrow Y$ be a function with $X \neq \emptyset$. Show that f is injective if and only if it admits a left inverse.
- b) Let $f : X \rightarrow Y$ be a function. Show that f is bijective if and only if it admits an inverse